LINE EQUATIONS

prepared by:

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GRADING SYSTEM

FINAL GRADE

30%

PRELIM

30%

MIDTERM

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SEATWORK

0

ABSENT

3

45-min LATE

7 max

15-min LATE

TOPIC OUTLINE

Equation of a Line

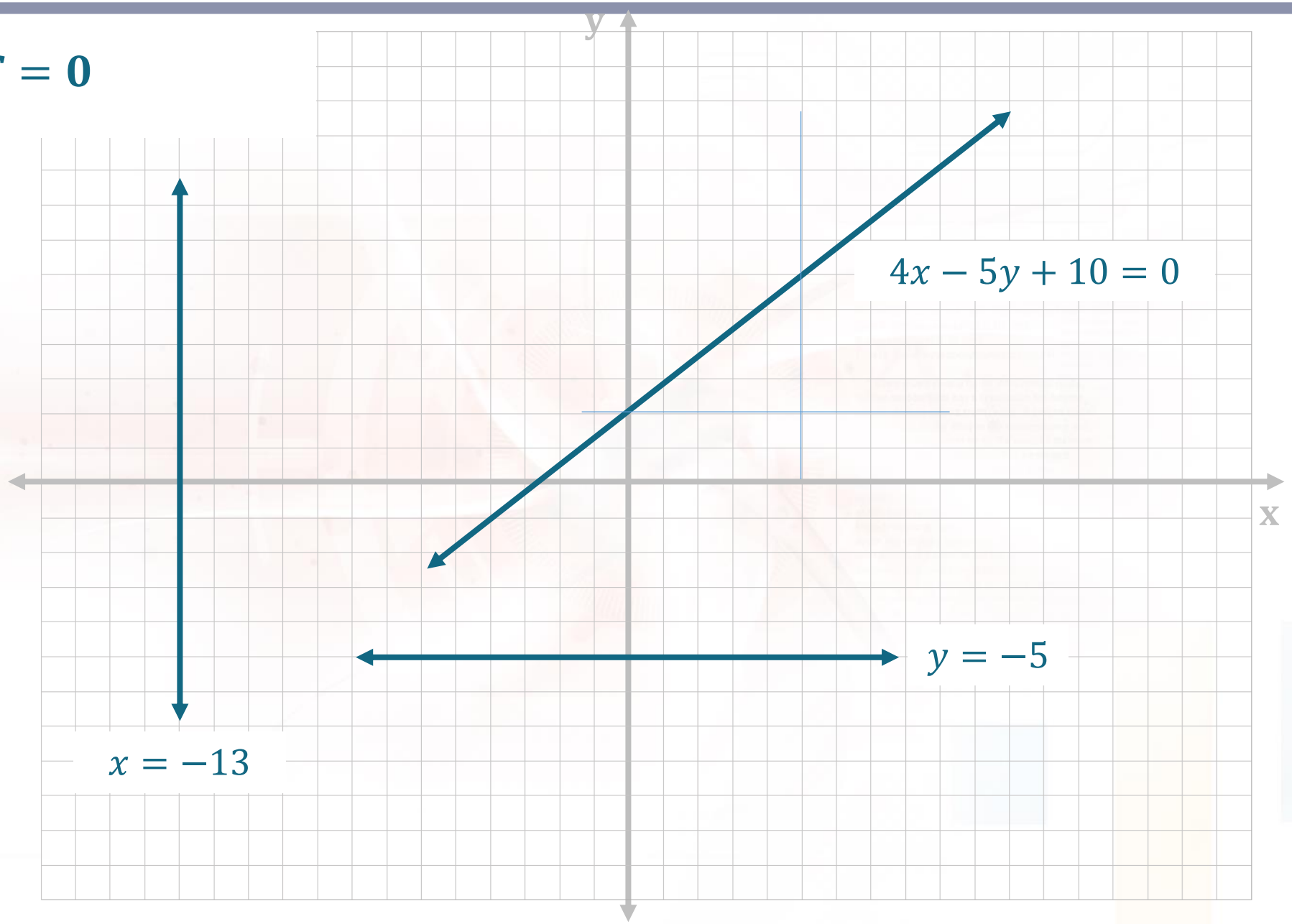
- General Equation
- Point Slope Form
- Slope Intercept Form
- Intercept Form
- Normal Form

Parallel and Perpendicular Lines

EQUATIONS OF A LINE

GENERAL EQUATION

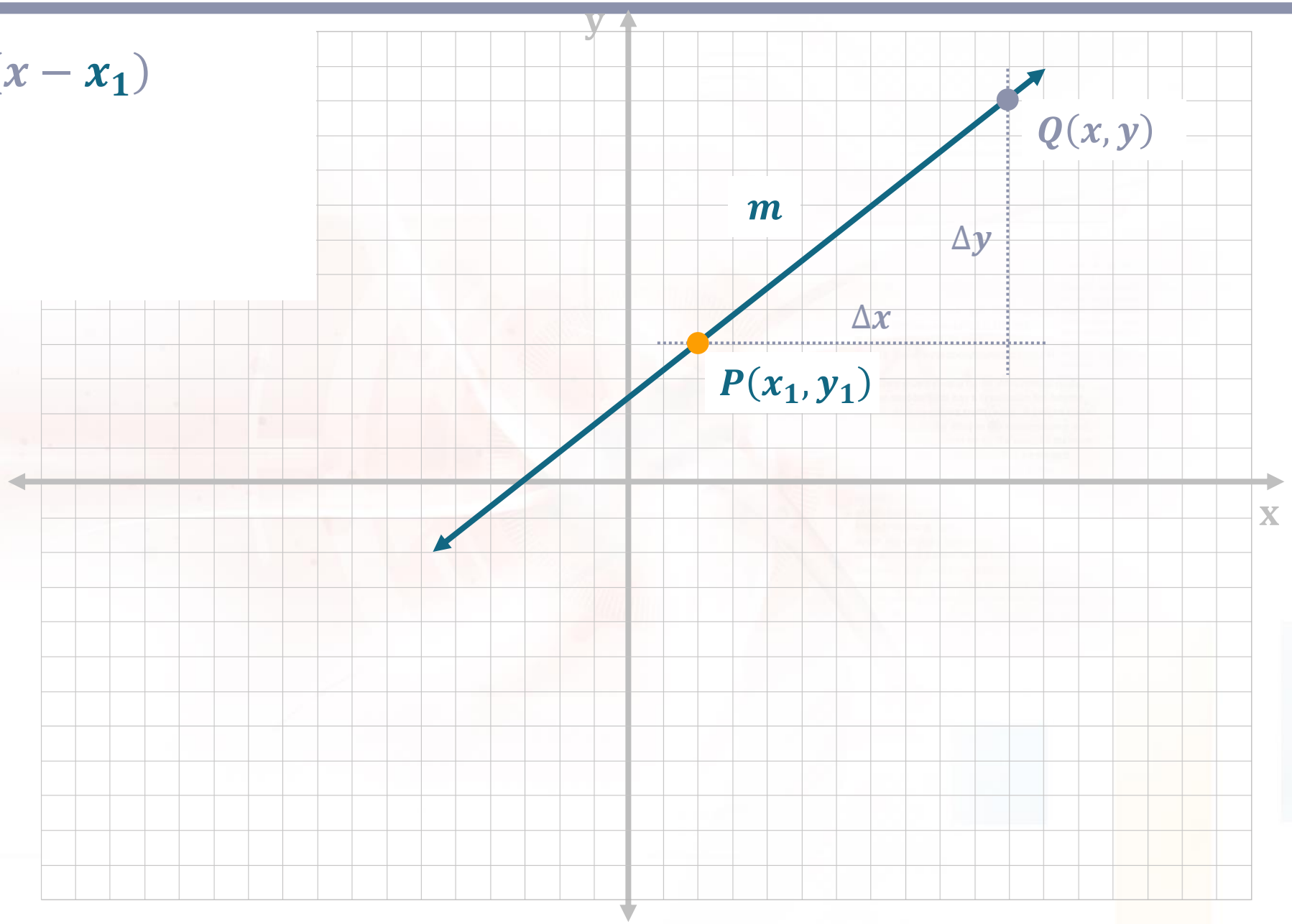
$$Ax + By + C = 0$$



POINT SLOPE FORM

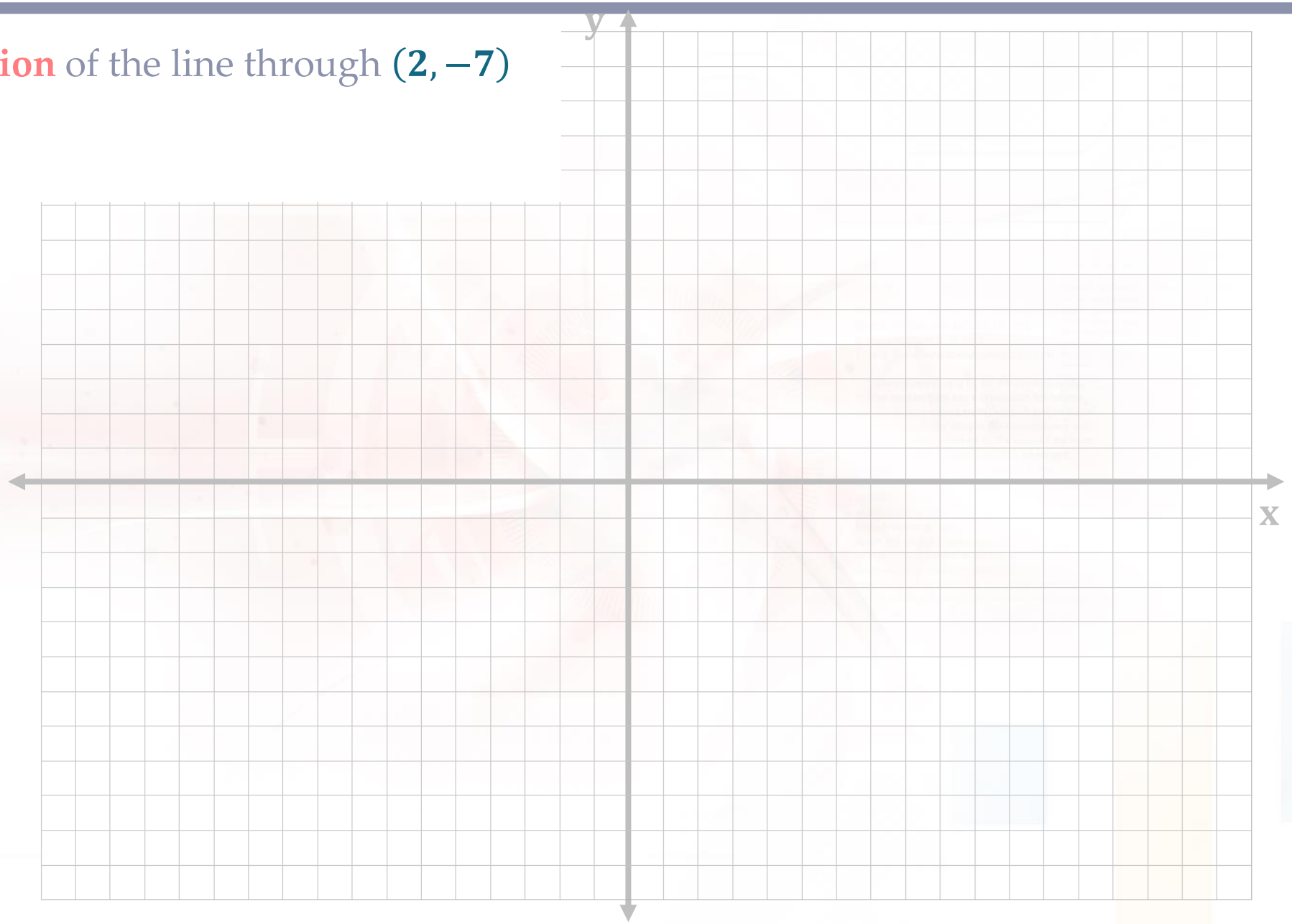
$$y - y_1 = m(x - x_1)$$

$$m = \frac{\Delta y}{\Delta x}$$



Find the **equation** of the line through $(2, -7)$

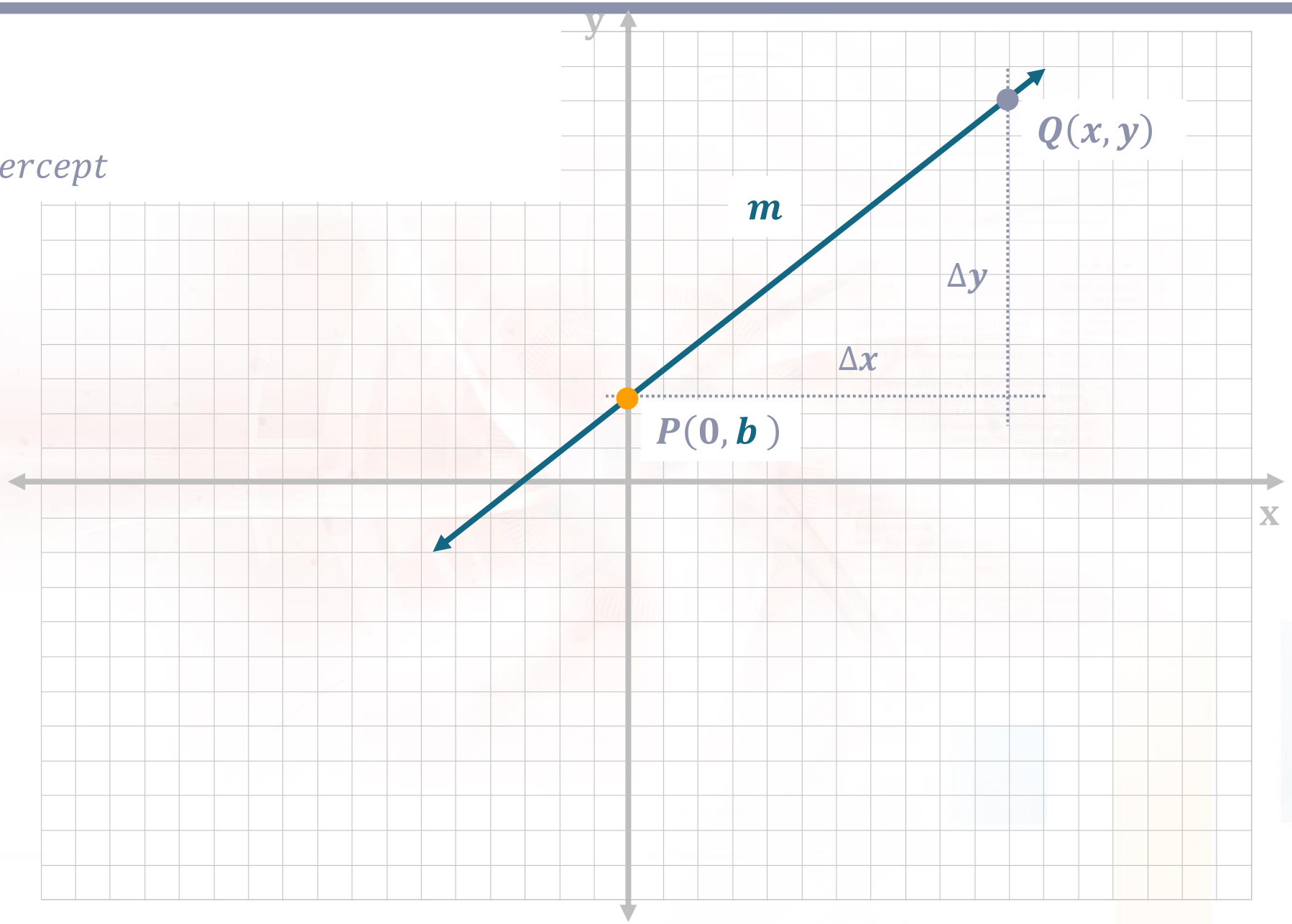
with **slope** $\frac{2}{5}$



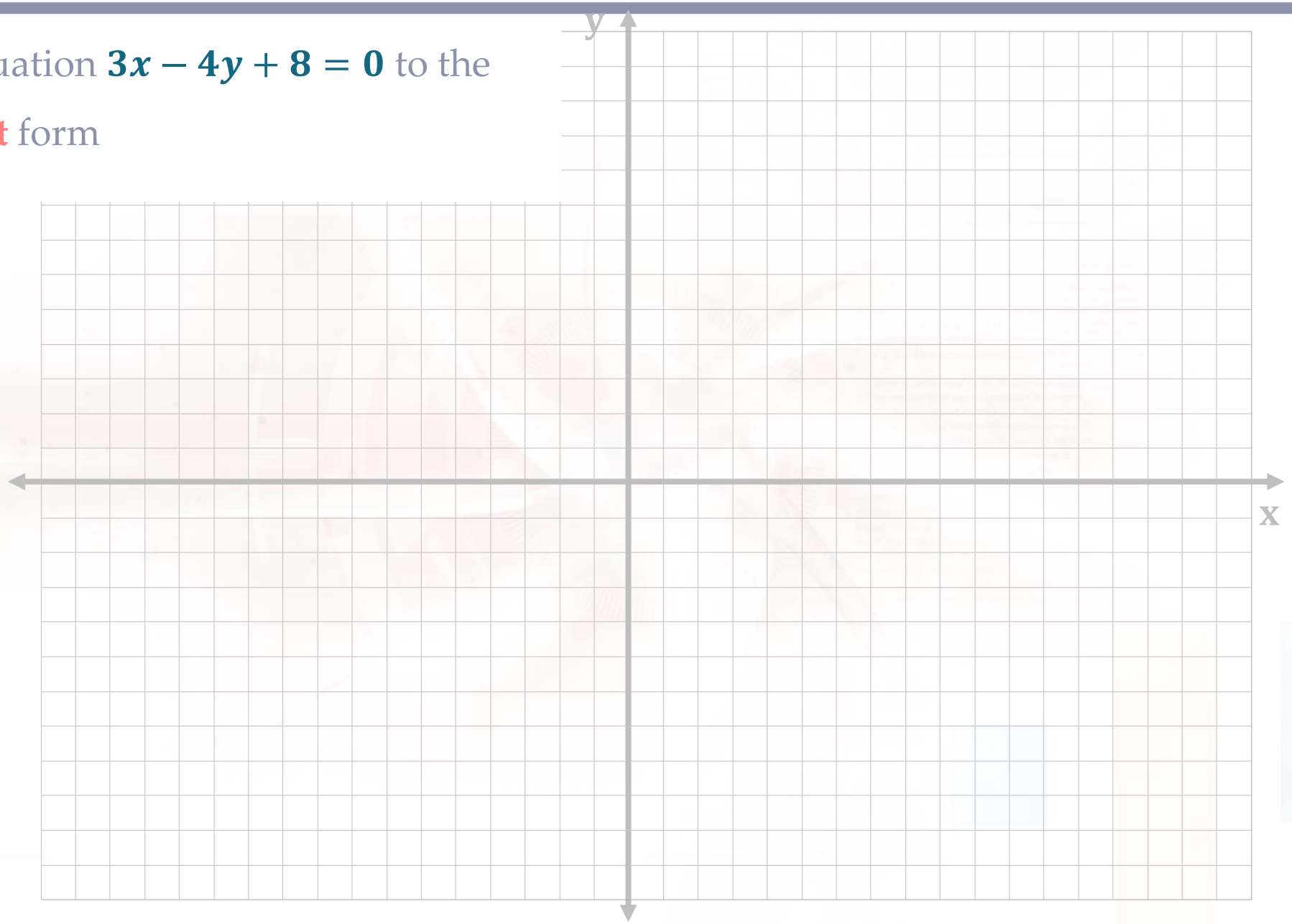
SLOPE INTERCEPT FORM

$$y = mx + b$$

b is the $y - intercept$



Reduce the equation $\mathbf{3x - 4y + 8 = 0}$ to the **slope intercept** form

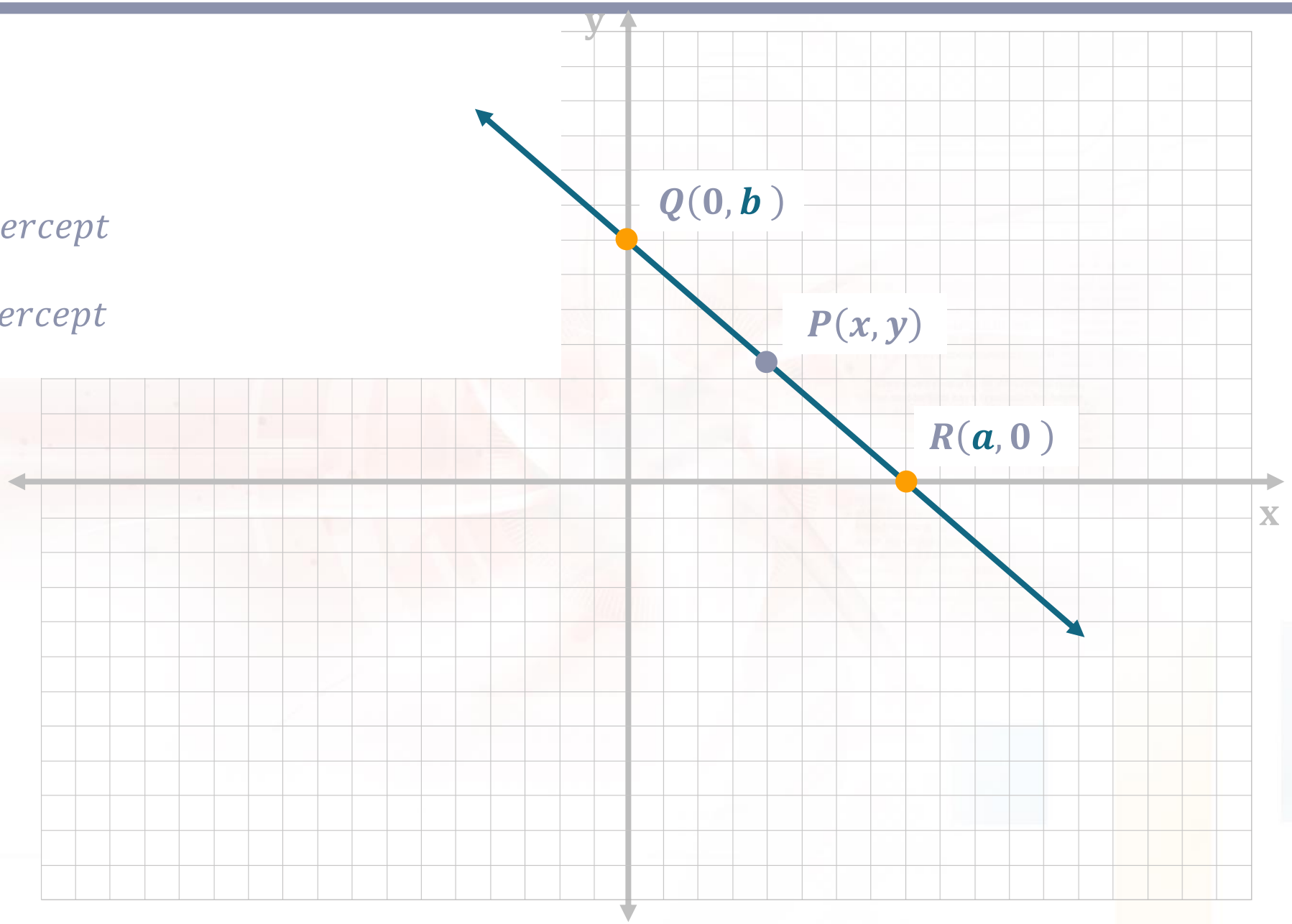


INTERCEPT FORM

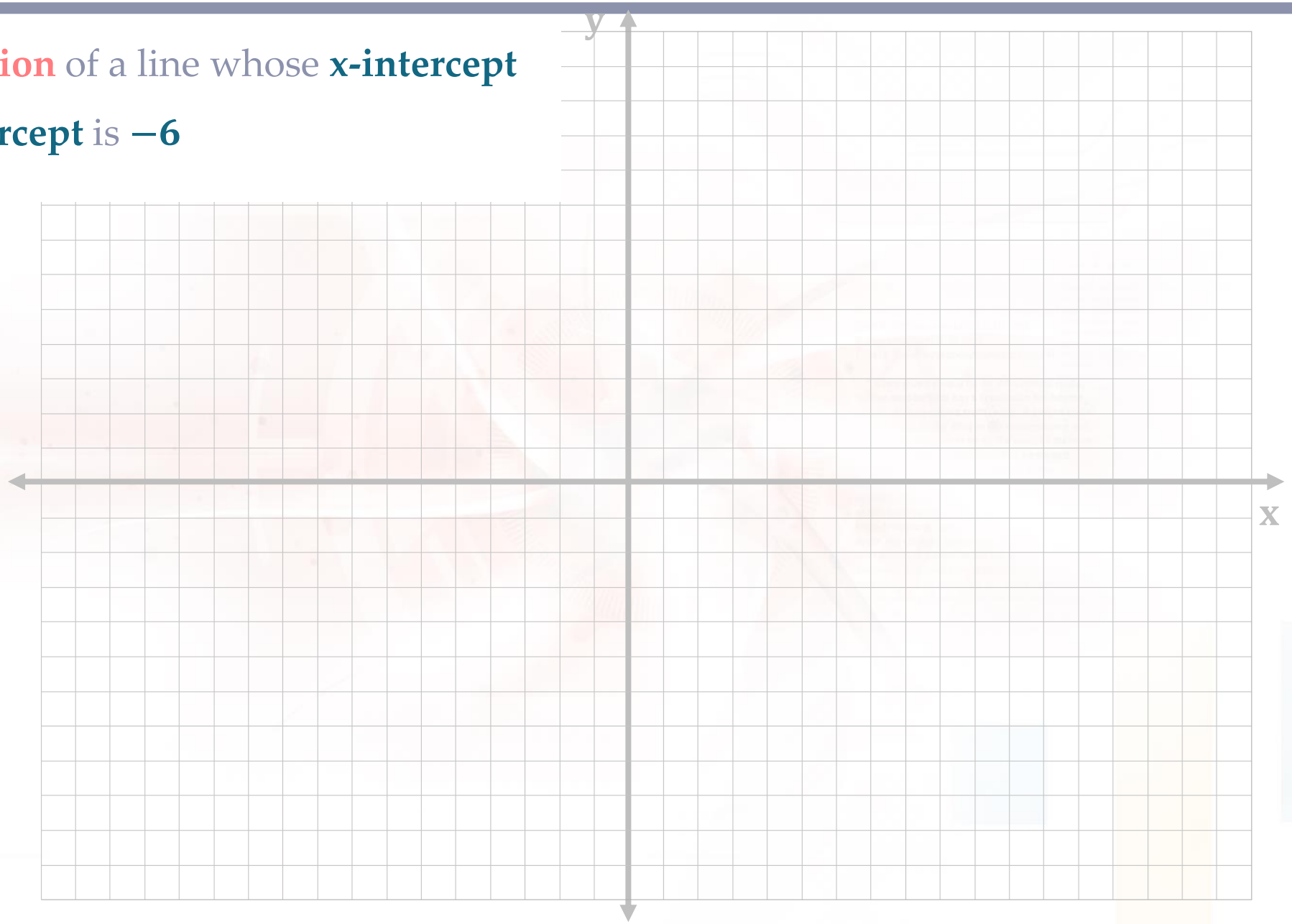
$$\frac{x}{a} + \frac{y}{b} = 1$$

a is the $x - intercept$

b is the $y - intercept$

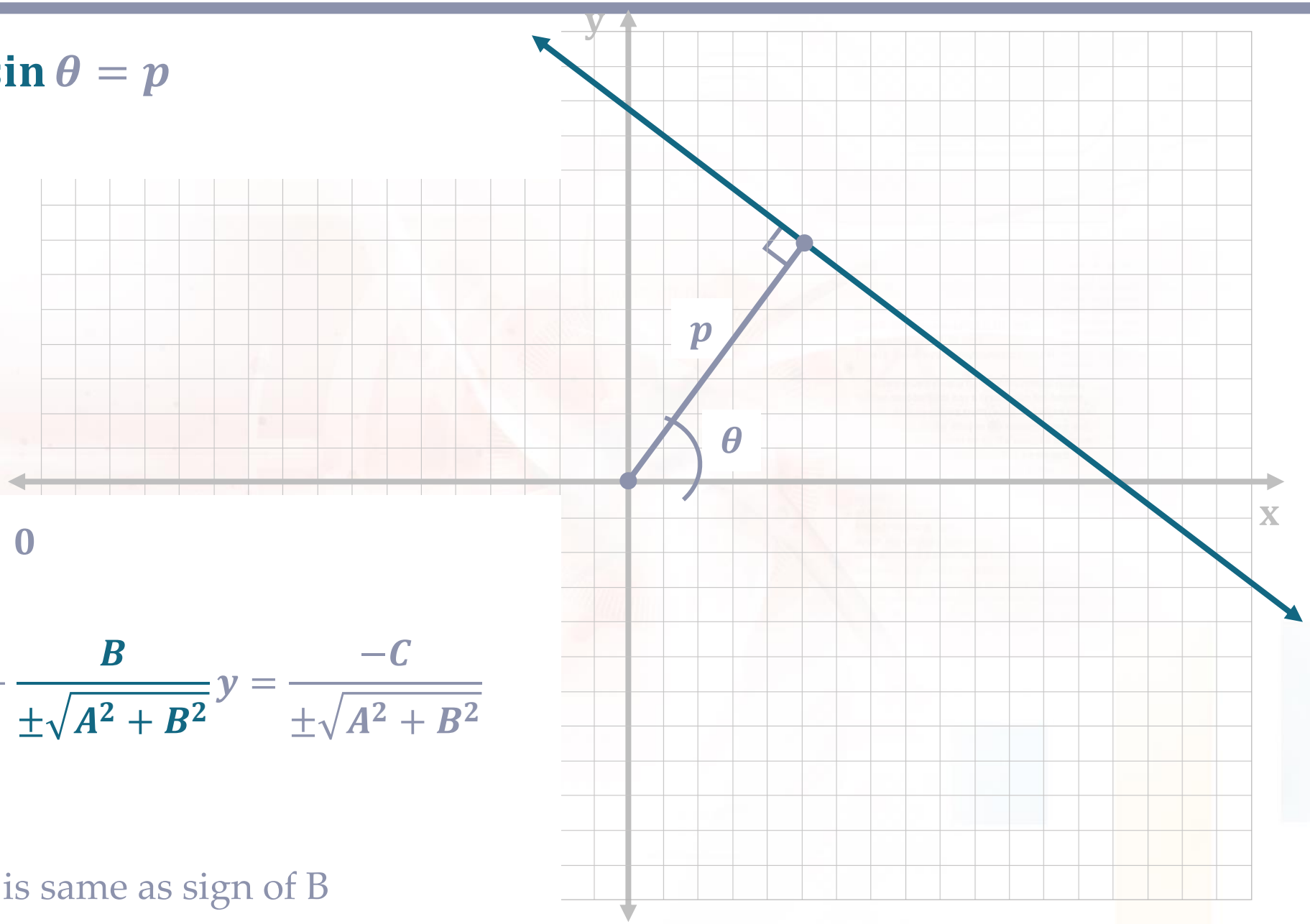


Find the **equation** of a line whose **x-intercept** is **4** and **y-intercept** is **−6**



NORMAL FORM

$$x\cos\theta + y\sin\theta = p$$



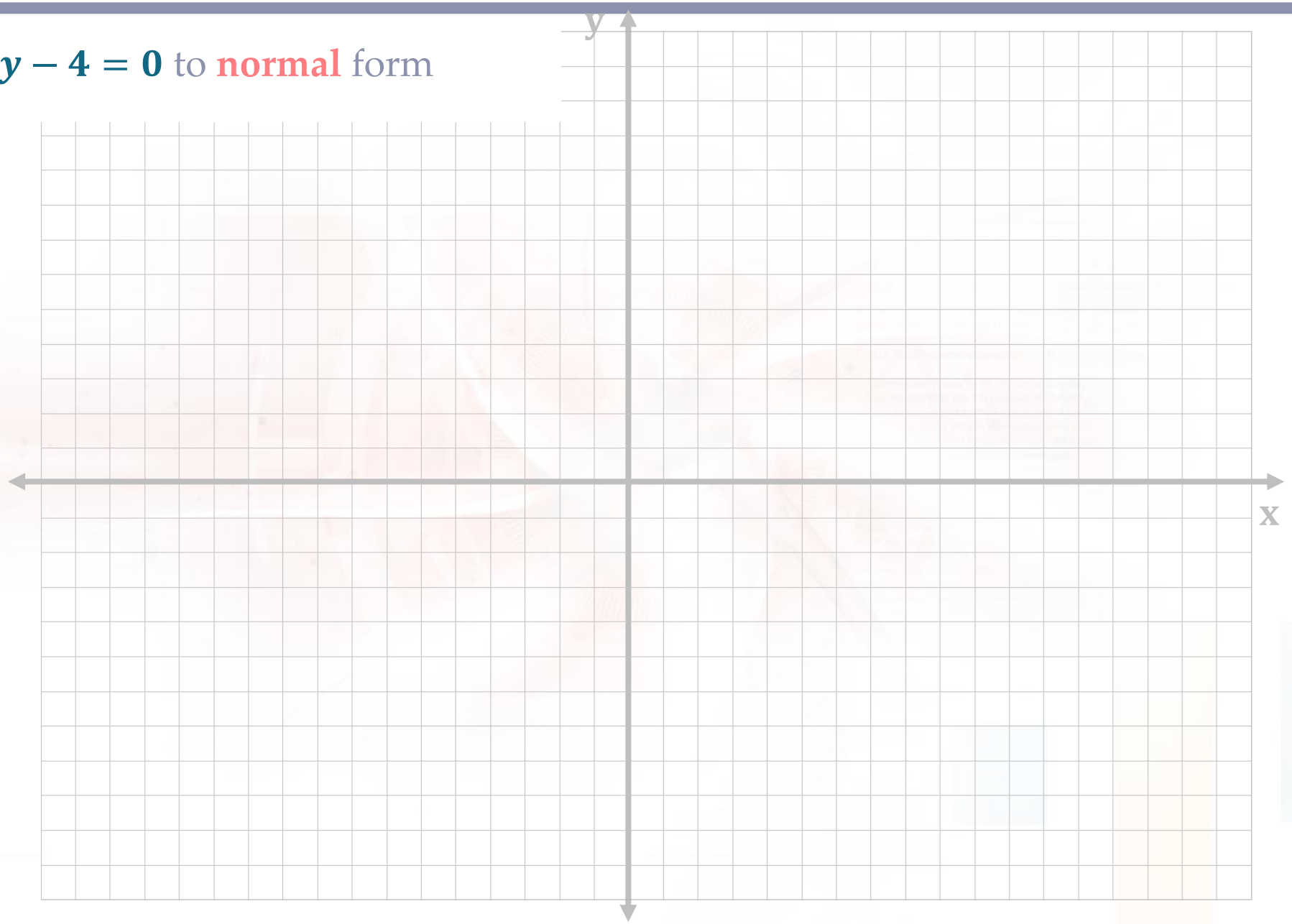
$$Ax + By + C = 0$$

$$\frac{A}{\pm\sqrt{A^2+B^2}}x + \frac{B}{\pm\sqrt{A^2+B^2}}y = \frac{-C}{\pm\sqrt{A^2+B^2}}$$

note:

Sign of radical is same as sign of B

Reduce $\mathbf{5x + 3y - 4 = 0}$ to **normal** form

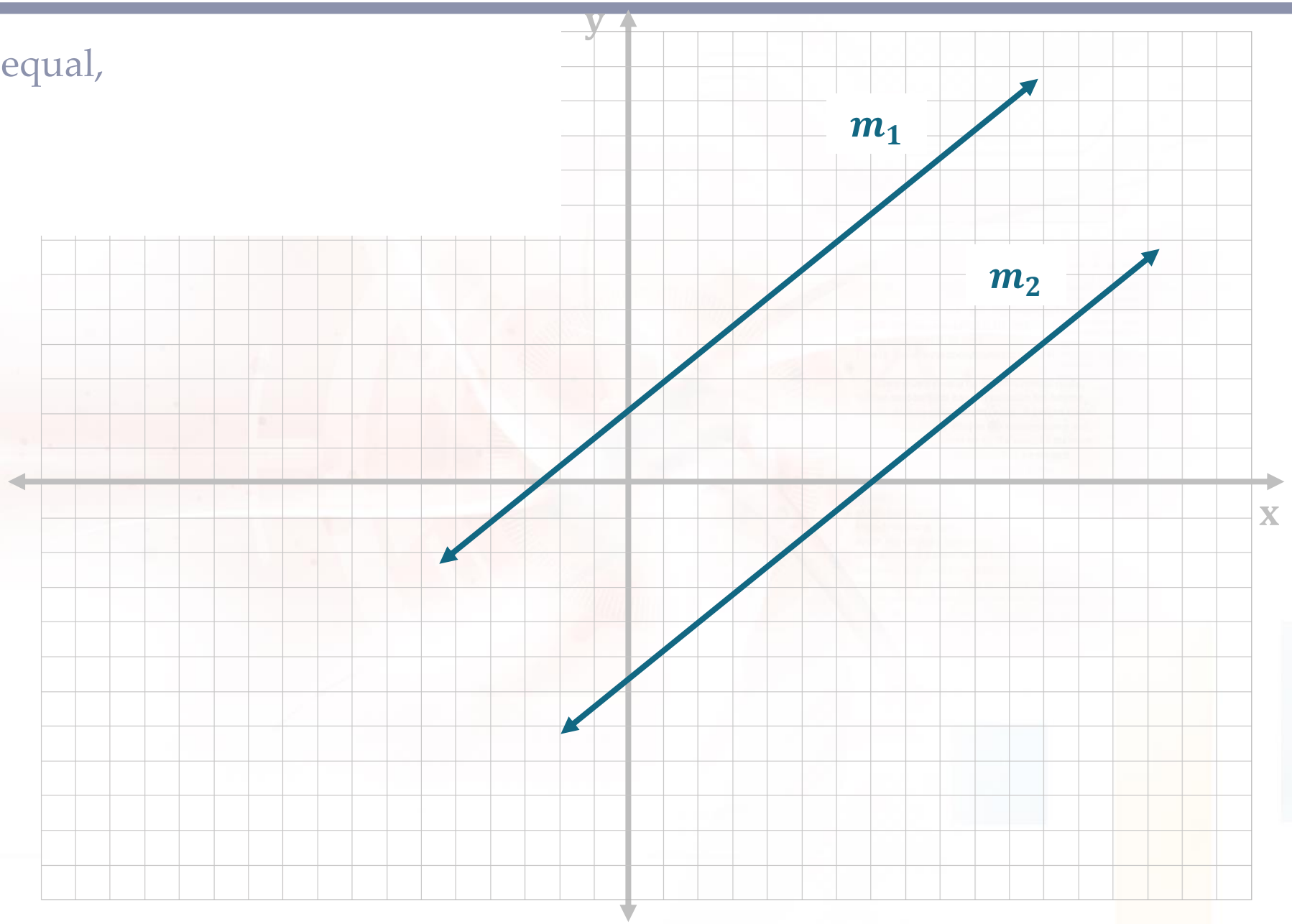


PARALLEL AND PERPENDICULAR LINES

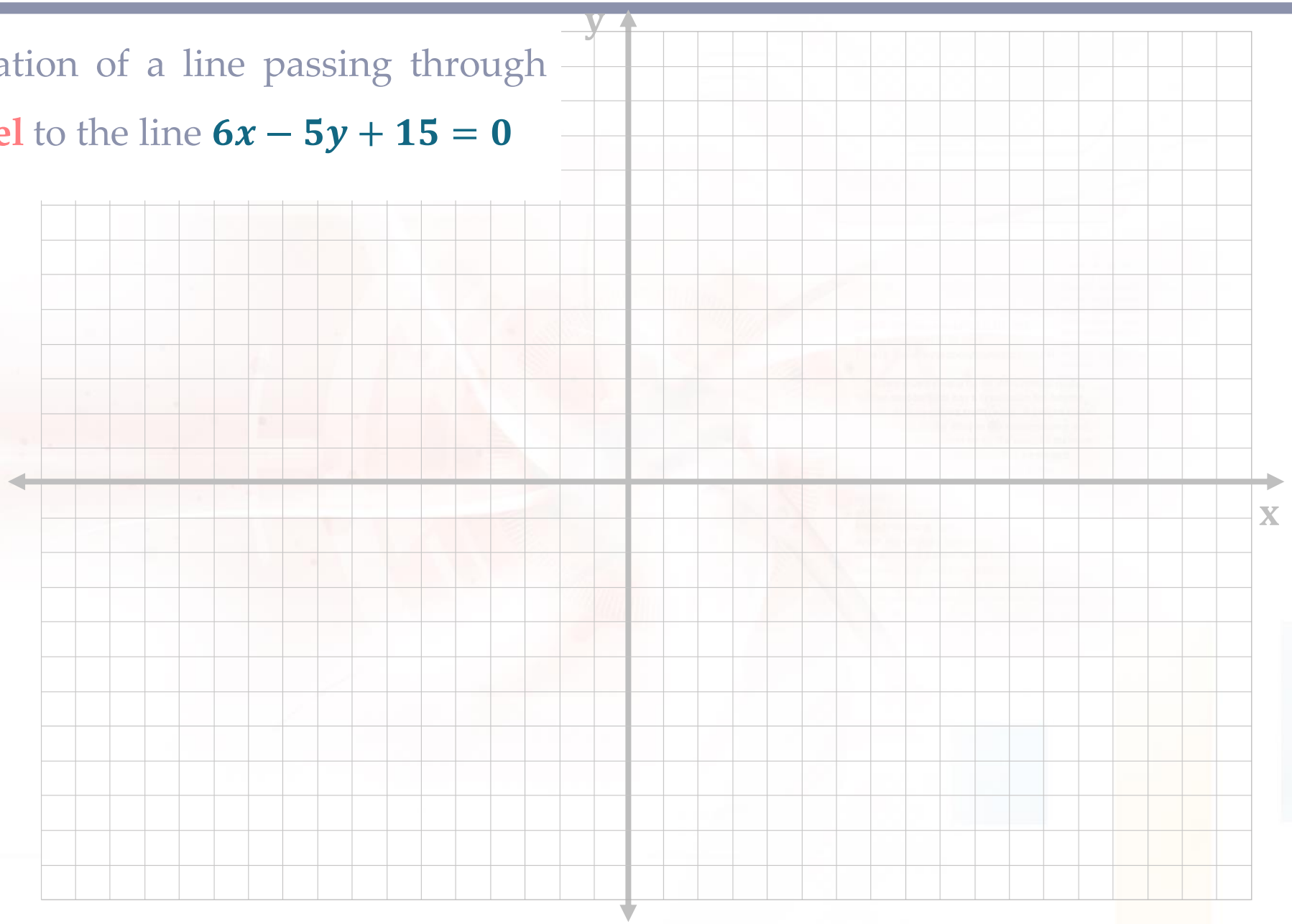
PARALLEL LINES

The slopes are equal,

$$m_1 = m_2$$

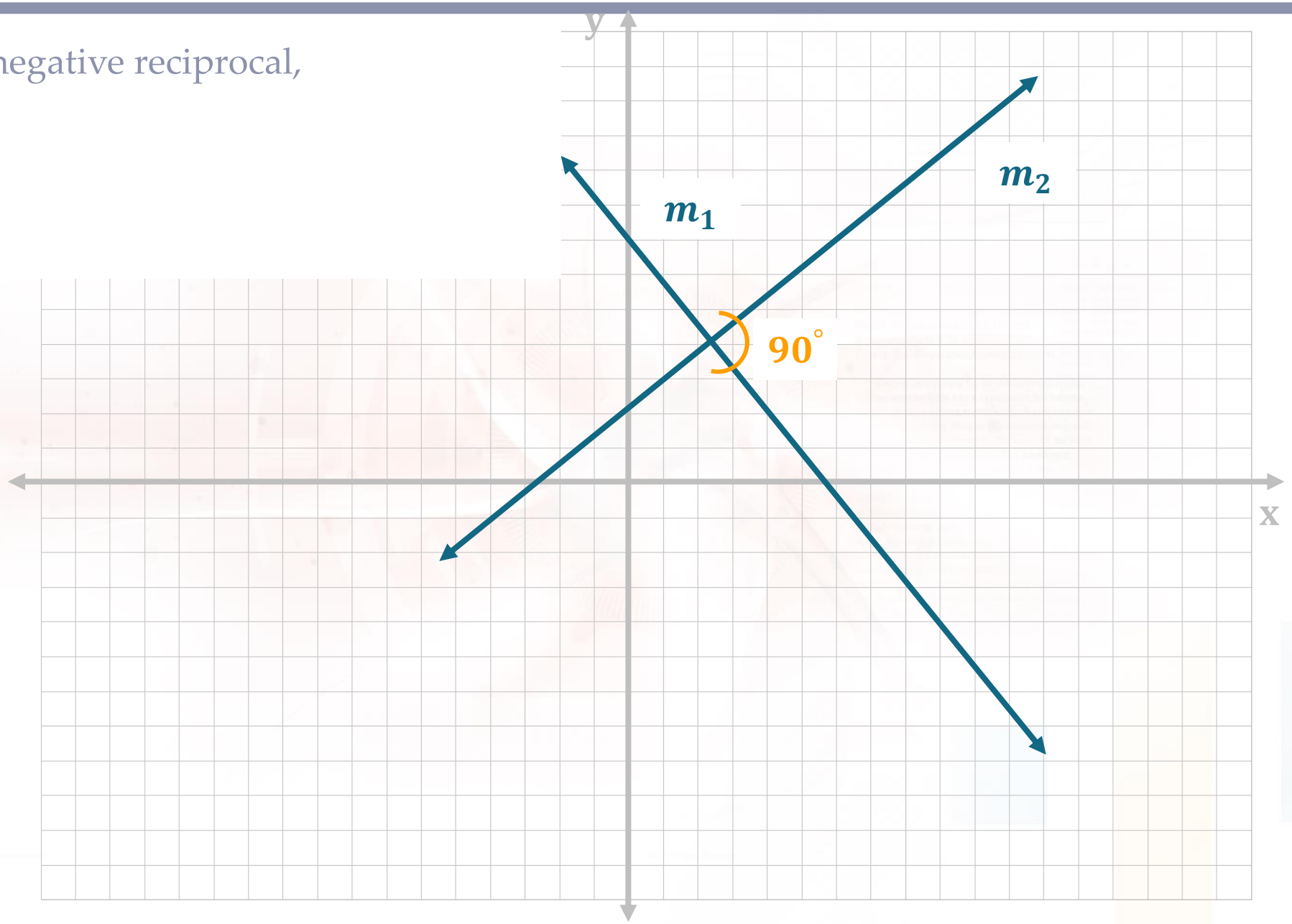


Write the equation of a line passing through $(-3, 8)$, **parallel** to the line $6x - 5y + 15 = 0$

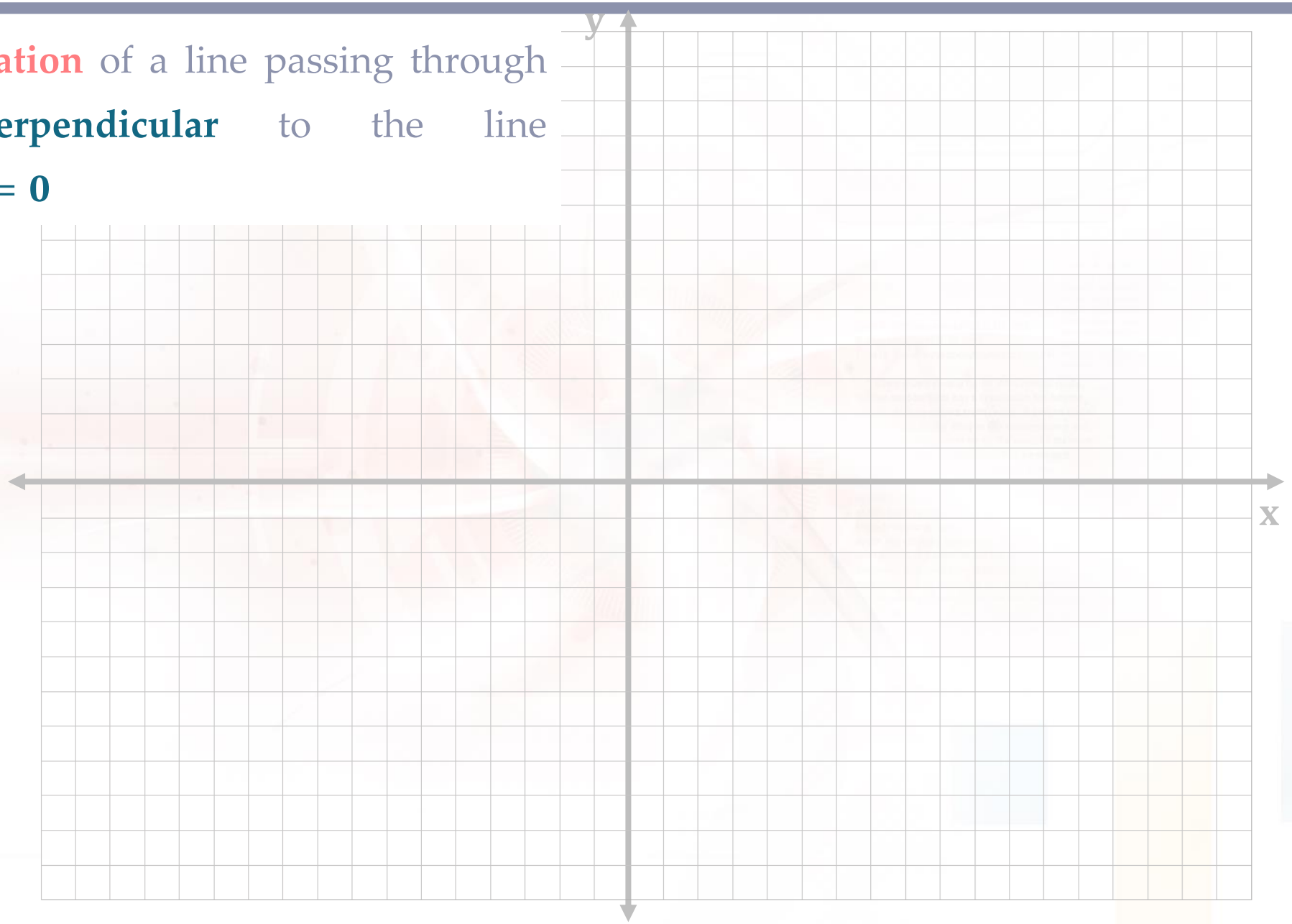


The slopes are negative reciprocal,

$$m_1 = -\frac{1}{m_2}$$

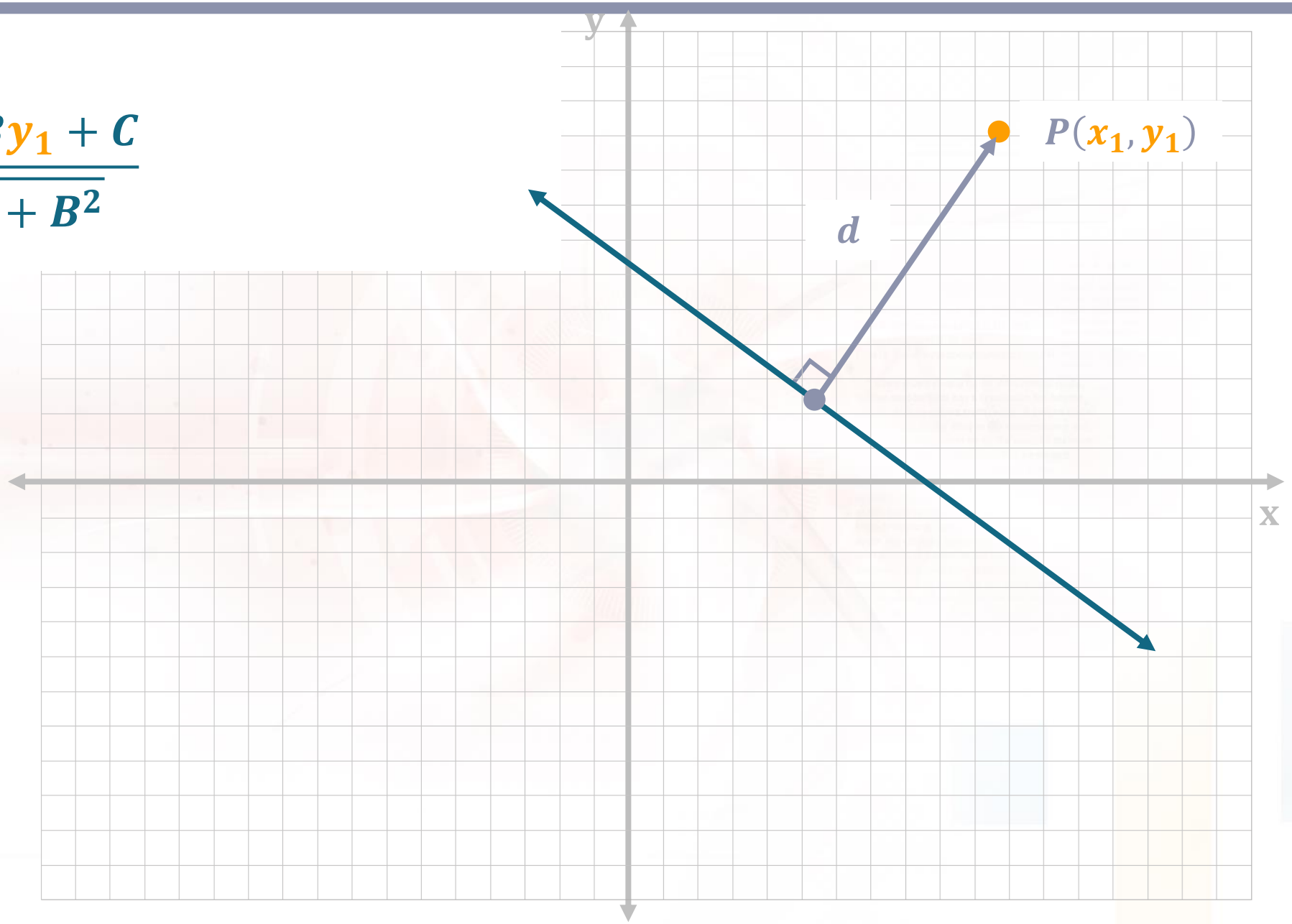


Write the **equation** of a line passing through $(-3, 8)$ **perpendicular** to the line $6x - 5y + 15 = 0$

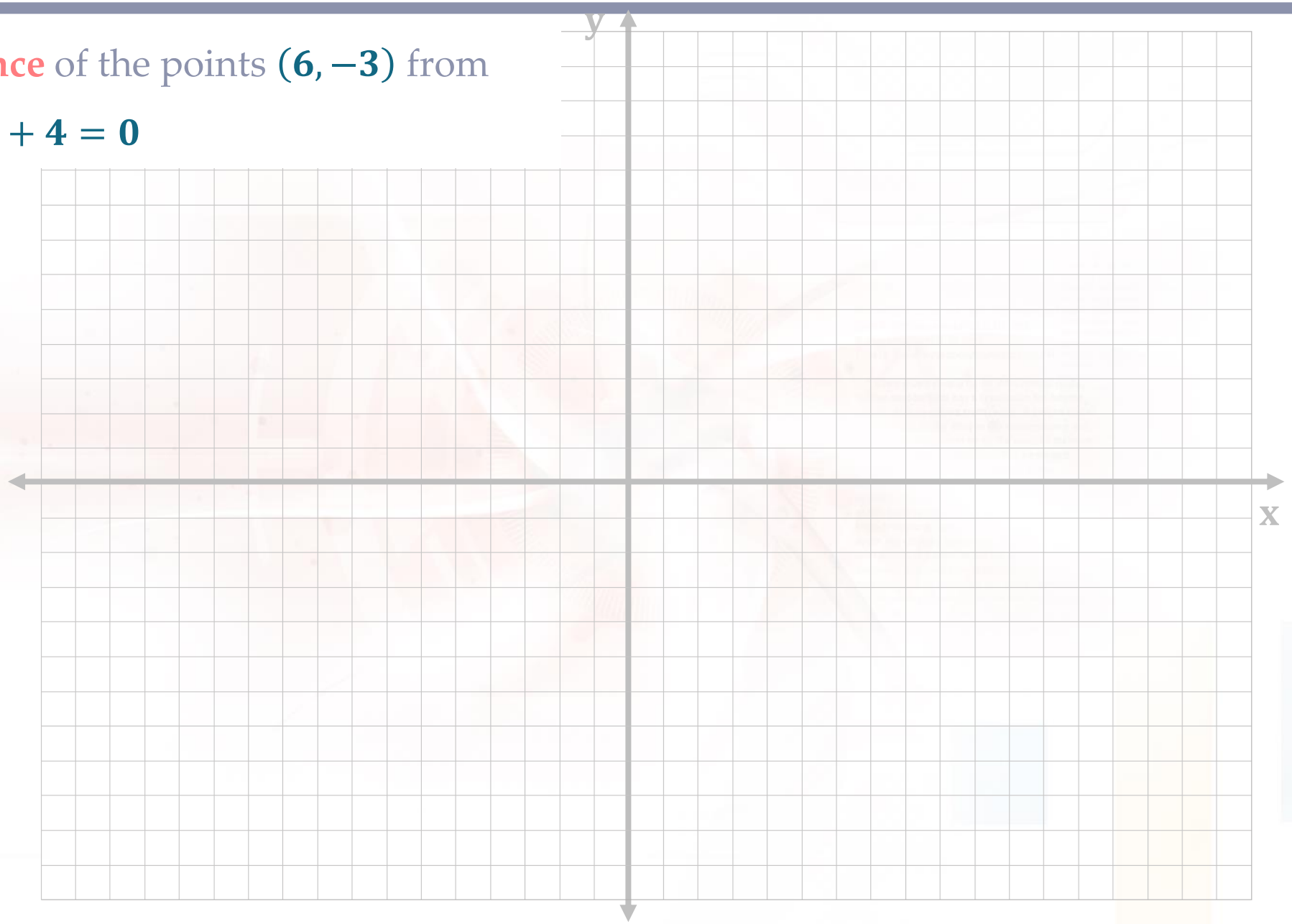


DISTANCE FROM A LINE TO A POINT

$$d = \frac{Ax_1 + By_1 + C}{\pm\sqrt{A^2 + B^2}}$$



Find the **distance** of the points $(\mathbf{6}, -\mathbf{3})$ from the line $\mathbf{2x - y + 4 = 0}$



SEATWORK

1. Find a **general form** of an equation of the line through the point $\boldsymbol{A(5,-3)}$ with the slope $\boldsymbol{m=-4}$.

2. Find the **equation** of the line passing through $\boldsymbol{(2,2)}$ **perpendicular** to $\boldsymbol{4x-2y=6}$.

3. The following table gives the cost (in thousands of dollars) for a 30-second television advertisement during the Super Bowl for various years. Determine a **line** in the form, where **x** is the **year** and **y** is the **cost** that models the data.

hint: slope intercept form

Year	Cost
1986	**550**
1996	**1085**
2001	**2100**
2005	**2400**